

Industrial Internship Report on Student Performance Prediction Using Machine Learning with Python

Prepared by
BALAJISANKRAVEL A

Executive Summary

This report presents the work completed during the **Winter Internship 2026** under **upskill Campus** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided an opportunity to gain practical knowledge in **Python programming, data analysis, and Machine Learning** by developing a real-world project titled "**Student Performance Prediction Using Machine Learning with Python.**"

The objective of the project was to develop a machine learning model capable of predicting a student's academic performance based on key factors such as study hours, attendance, previous examination marks, assignment scores, quiz scores, and internal assessment marks. By analyzing these factors, the system helps identify students who may require additional academic support, enabling educators to make informed decisions and improve learning outcomes.

The project was implemented using **Python** and popular machine learning libraries including **Pandas, NumPy, Scikit-learn, Matplotlib, and Joblib**. The development process involved data collection, data preprocessing, exploratory data analysis, model training, performance evaluation, and prediction. A **Random Forest Classifier** was used to train the model due to its high accuracy and reliability. The trained model was evaluated using standard performance metrics such as accuracy, confusion matrix, and classification report before being saved for future predictions. Overall, this internship was a valuable learning experience that enhanced both my technical and professional skills. It provided practical exposure to machine learning applications in the education domain and prepared me for future opportunities in software development, data science, and artificial intelligence. The successful completion of this project demonstrates my ability to develop data-driven solutions using Python and machine learning techniques.

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1 Preface

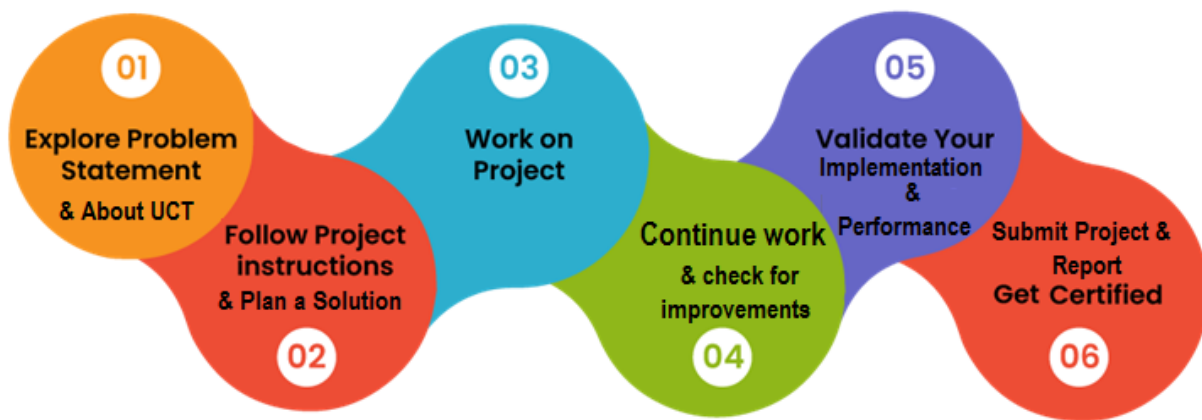
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



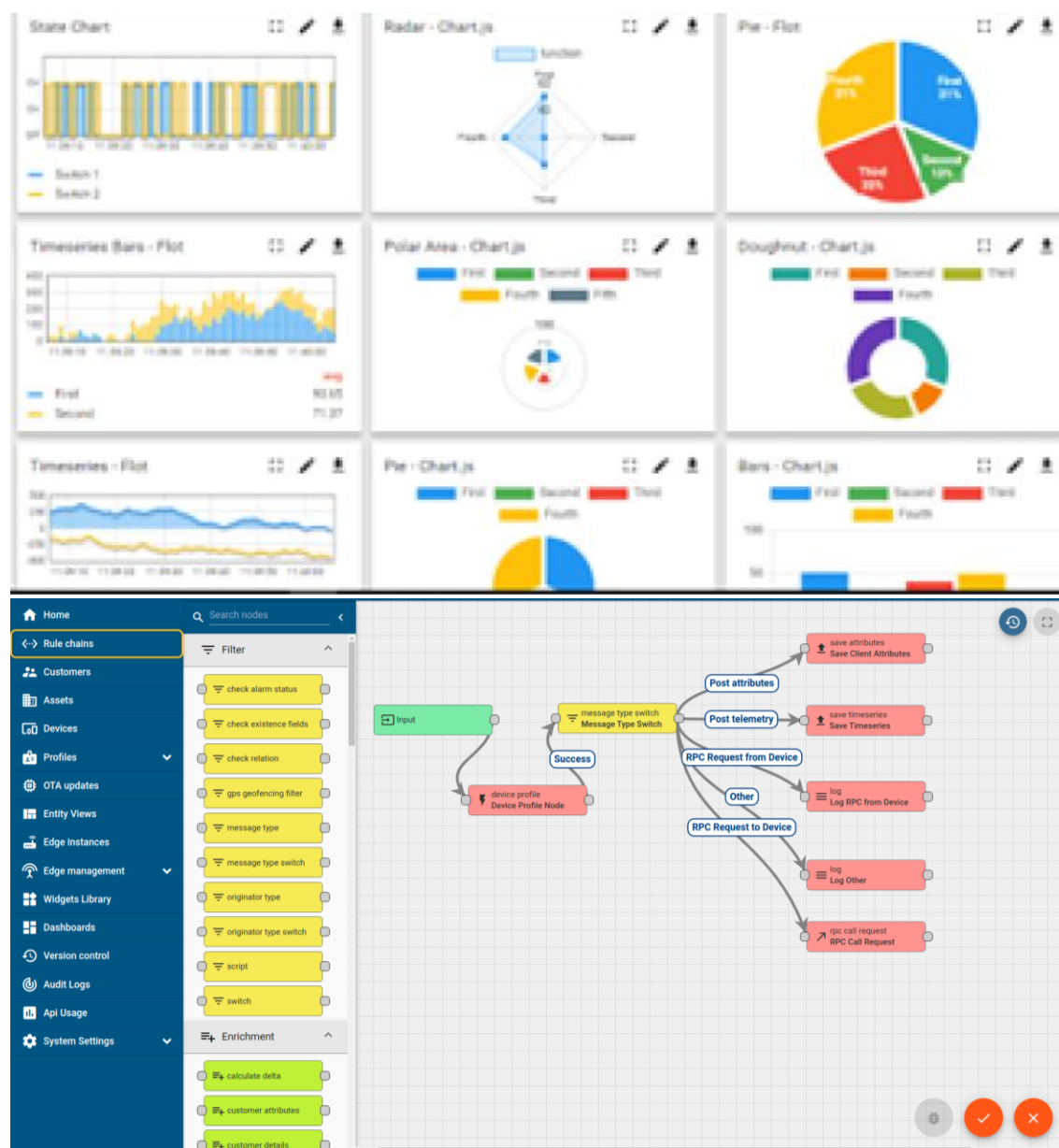
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



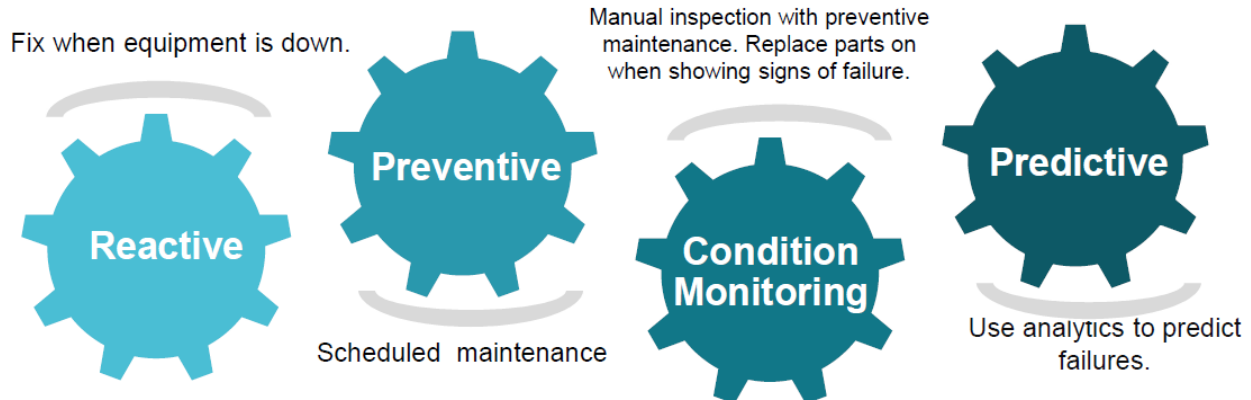


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

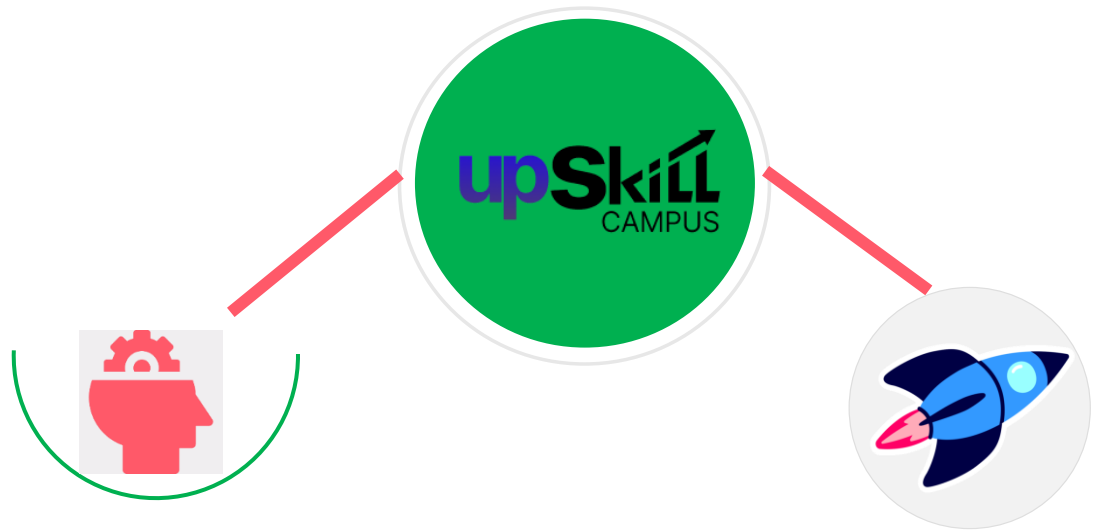
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

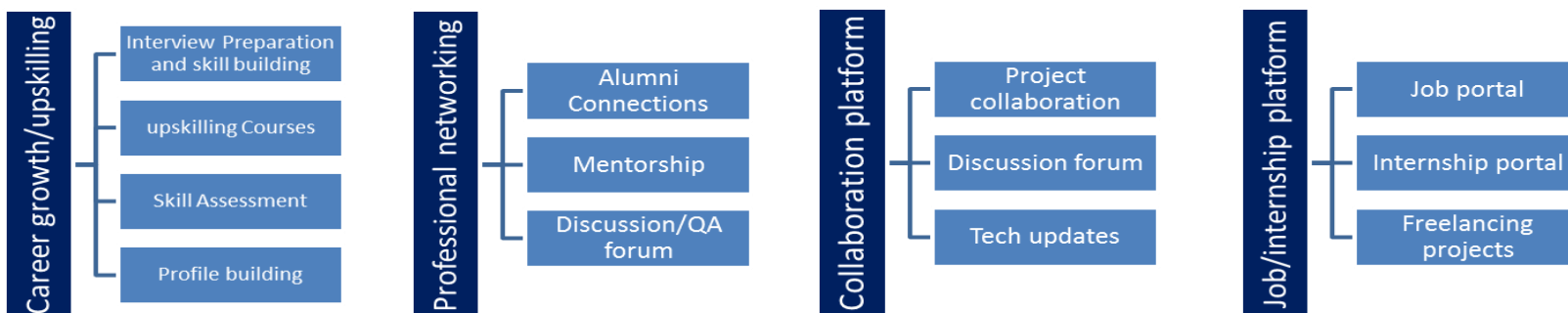
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd Edition, O'Reilly Media, 2022.
- [2] Wes McKinney, *Python for Data Analysis*, 3rd Edition, O'Reilly Media, 2022.
- [3] Scikit-learn Developers, *Scikit-learn Documentation*. Available at: <https://scikit-learn.org/>

2.6 Glossary

Terms	Acronym
AI	Artificial Intelligence
ML	Machine Learning
Dataset	Collection of data used for analysis
Linear Regression	Machine Learning algorithm used for prediction
Prediction	Estimating future outcomes using data

3 Problem Statement

Educational institutions generate a large amount of student academic data, including attendance records, examination marks, assignment scores, and quiz results. However, this data is often used only for maintaining records rather than for predicting student performance. As a result, educators may find it difficult to identify students who are at risk of poor academic performance before final examinations.

The lack of an intelligent prediction system makes it challenging to provide timely guidance and personalized support to students. Traditional methods of evaluating student performance are usually based only on final examination results, which may be too late for effective intervention.

The objective of this project is to develop a **Student Performance Prediction System** using **Machine Learning with Python**. The system analyzes important academic factors such as **study hours, attendance, previous marks, assignment scores, quiz scores, and internal assessment marks** to predict whether a student is likely to **pass or fail**.

The project uses machine learning algorithms to identify patterns in historical student data and generate accurate predictions. The trained model can assist teachers and educational institutions in recognizing students who may require additional academic support, enabling early intervention and improved learning outcomes.

4 Existing and Proposed solution

Traditional methods of evaluating student performance primarily rely on manual assessment of examination scores, attendance records, and assignment marks. Teachers analyze student performance at the end of each semester, making it difficult to identify academically weak students at an early stage. These methods are often time-consuming, less efficient, and do not provide predictive insights for future performance.

Several existing systems focus only on storing student records without using advanced analytics or machine learning techniques. As a result, institutions cannot effectively predict student outcomes or implement timely interventions.

Limitations of Existing Solution

- Manual analysis requires significant time and effort.
- Unable to predict future student performance.
- Limited use of historical academic data.
- Difficult to identify students at risk before final examinations.
- Lack of intelligent decision-making support.
- Reduced accuracy in forecasting academic outcomes.
- No automated prediction or recommendation system.

The proposed system is a **Student Performance Prediction System Using Machine Learning with Python**. It utilizes historical student data, including study hours, attendance, previous marks, assignment scores, quiz scores, and internal assessment marks, to predict whether a student is likely to **Pass** or **Fail**.

The system follows the complete machine learning workflow, including data collection, preprocessing, model training, evaluation, and prediction. A **Random Forest Classifier** is used as the prediction model due to its high accuracy and robustness.

The application is implemented in Python using libraries such as **Pandas**, **NumPy**, **Scikit-learn**, **Matplotlib**, and **Joblib**. After training, the model is saved and can be used repeatedly to predict the performance of new students without retraining.

4.1 Code submission (Github link) [Naveenjs5689/UpskillsCampus](https://github.com/Naveenjs5689/UpskillsCampus)

4.2 Report submission (Github link) : [Naveenjs5689/UpskillsCampus](https://github.com/Naveenjs5689/UpskillsCampus)

5 Proposed Design/ Model

The proposed system is a **Machine Learning-based Student Performance Prediction System** developed using **Python**. The system predicts whether a student is likely to **Pass** or **Fail** by analyzing important academic parameters such as study hours, attendance, previous marks, assignment scores, quiz scores, and internal assessment marks.

The system follows a structured machine learning workflow consisting of data collection, preprocessing, model training, evaluation, and prediction. Historical student data is used to train a **Random Forest Classifier**, which learns patterns from the dataset and predicts the performance of new students based on their academic information.

The proposed model is designed to improve the efficiency of academic performance analysis by providing accurate predictions that can assist educators in identifying students who may require additional academic support.

5.1 High Level Diagram

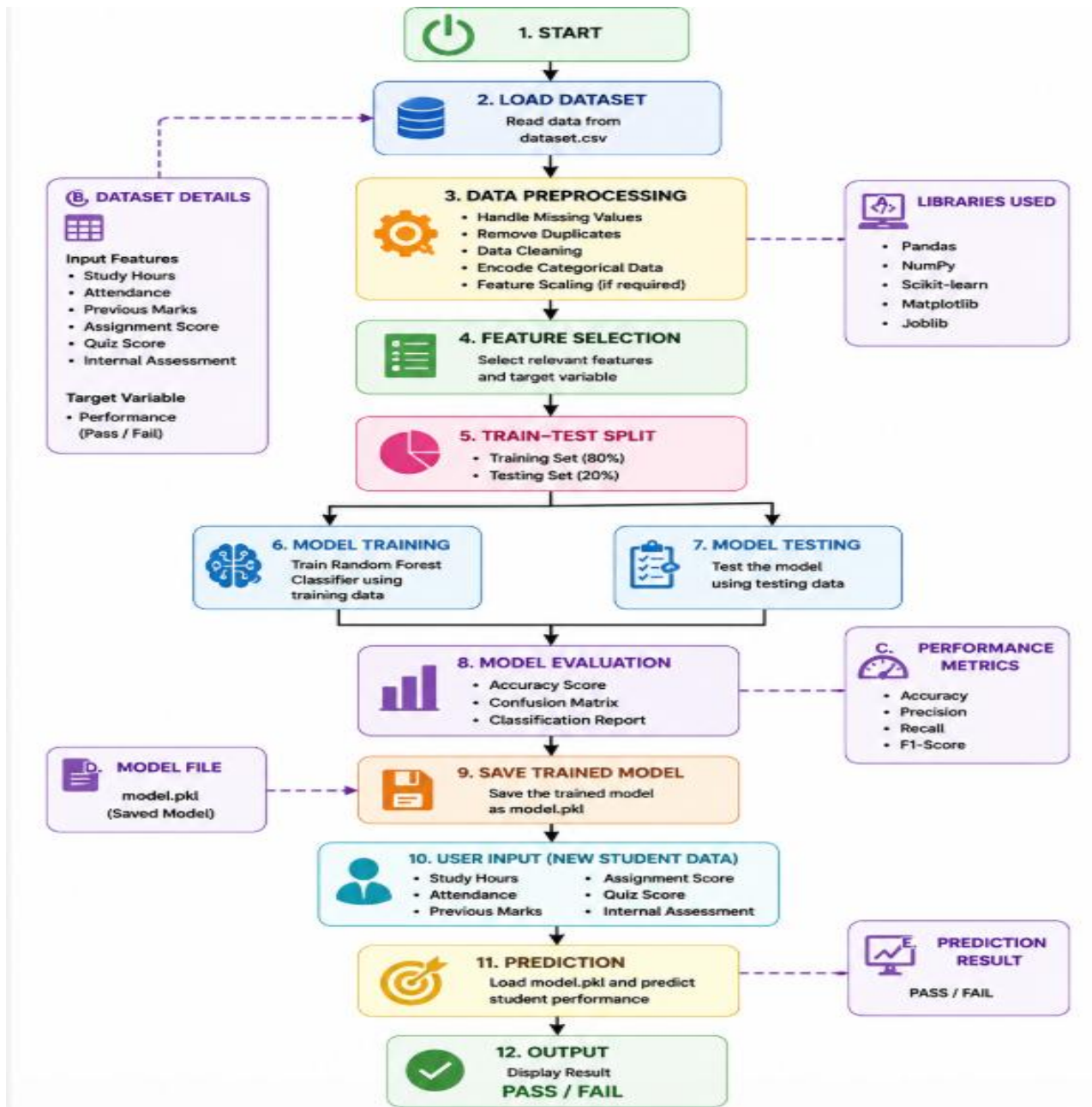
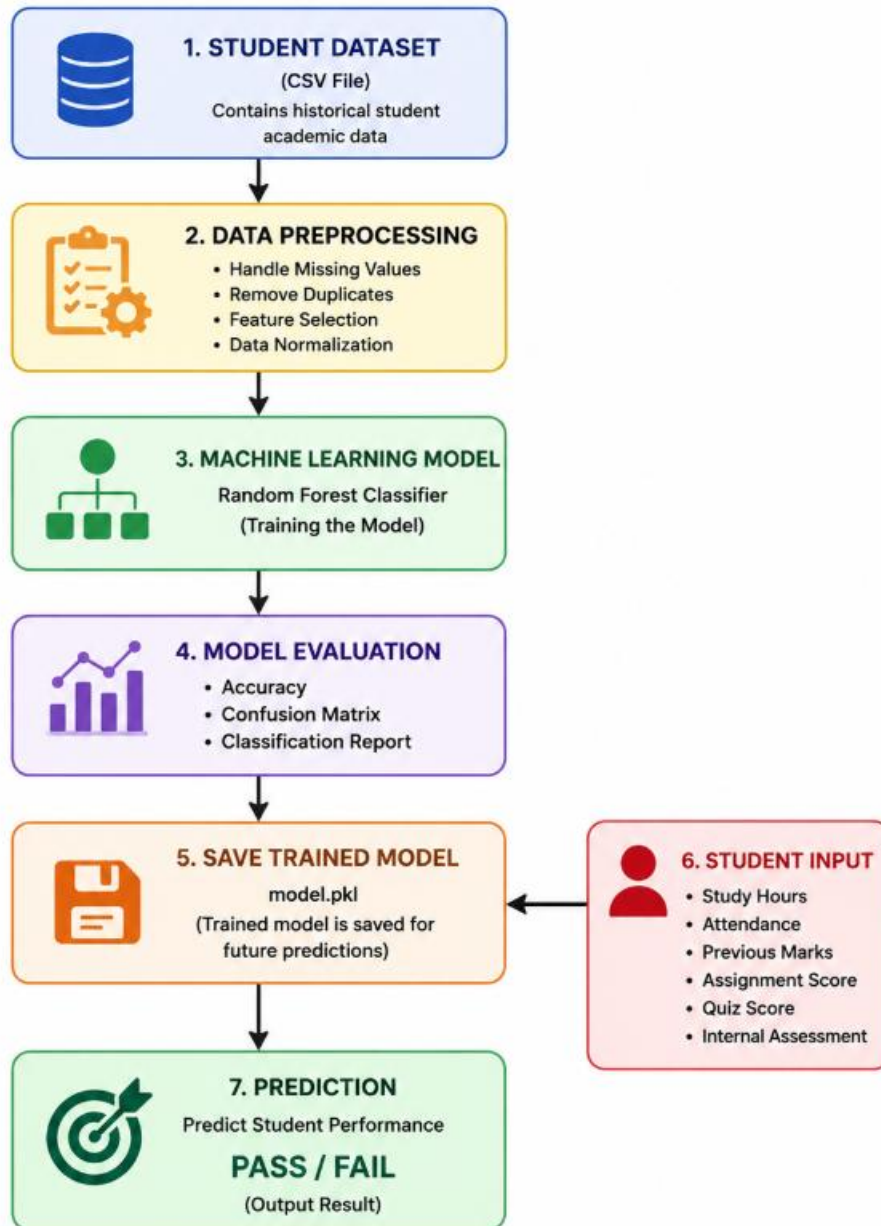
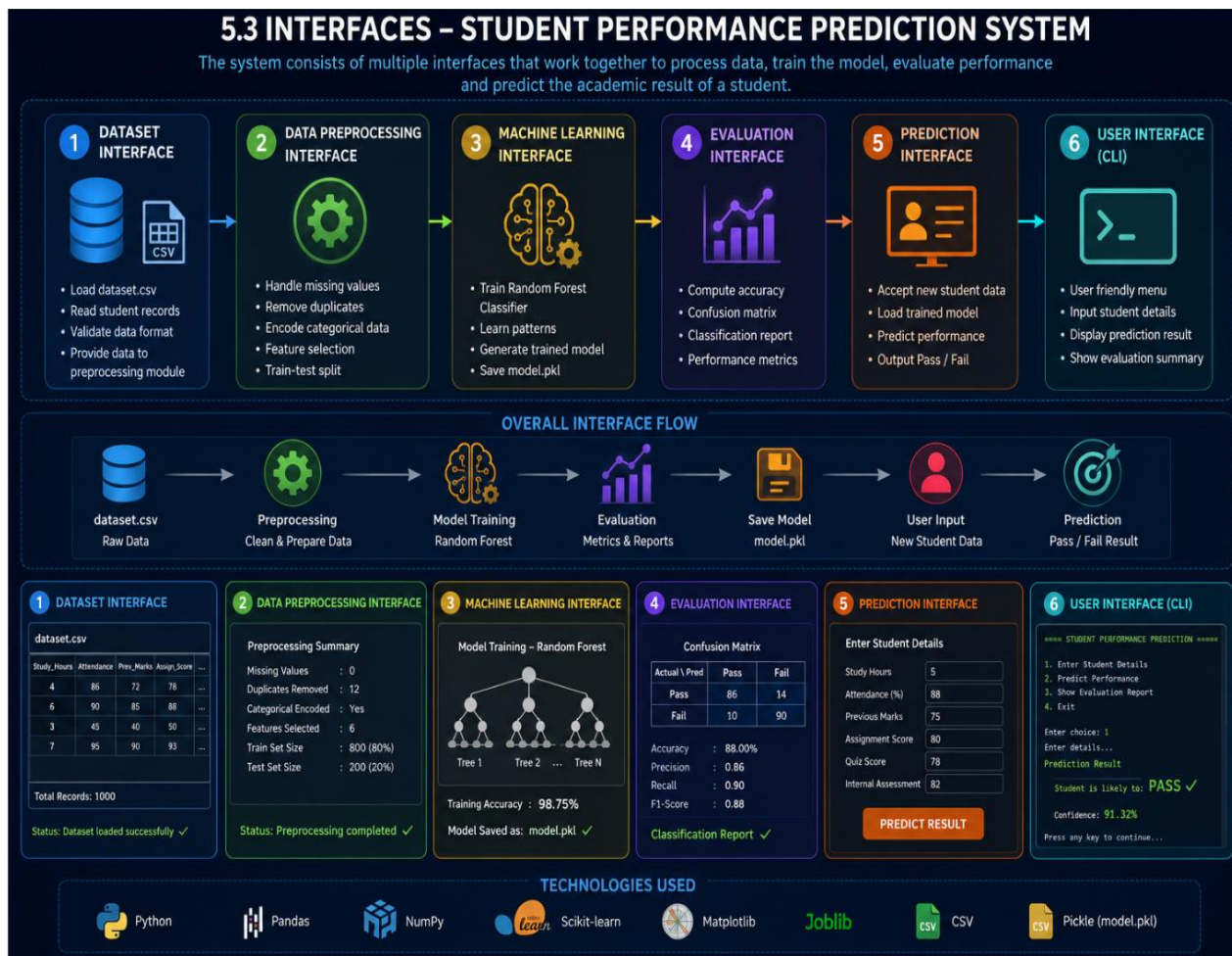


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



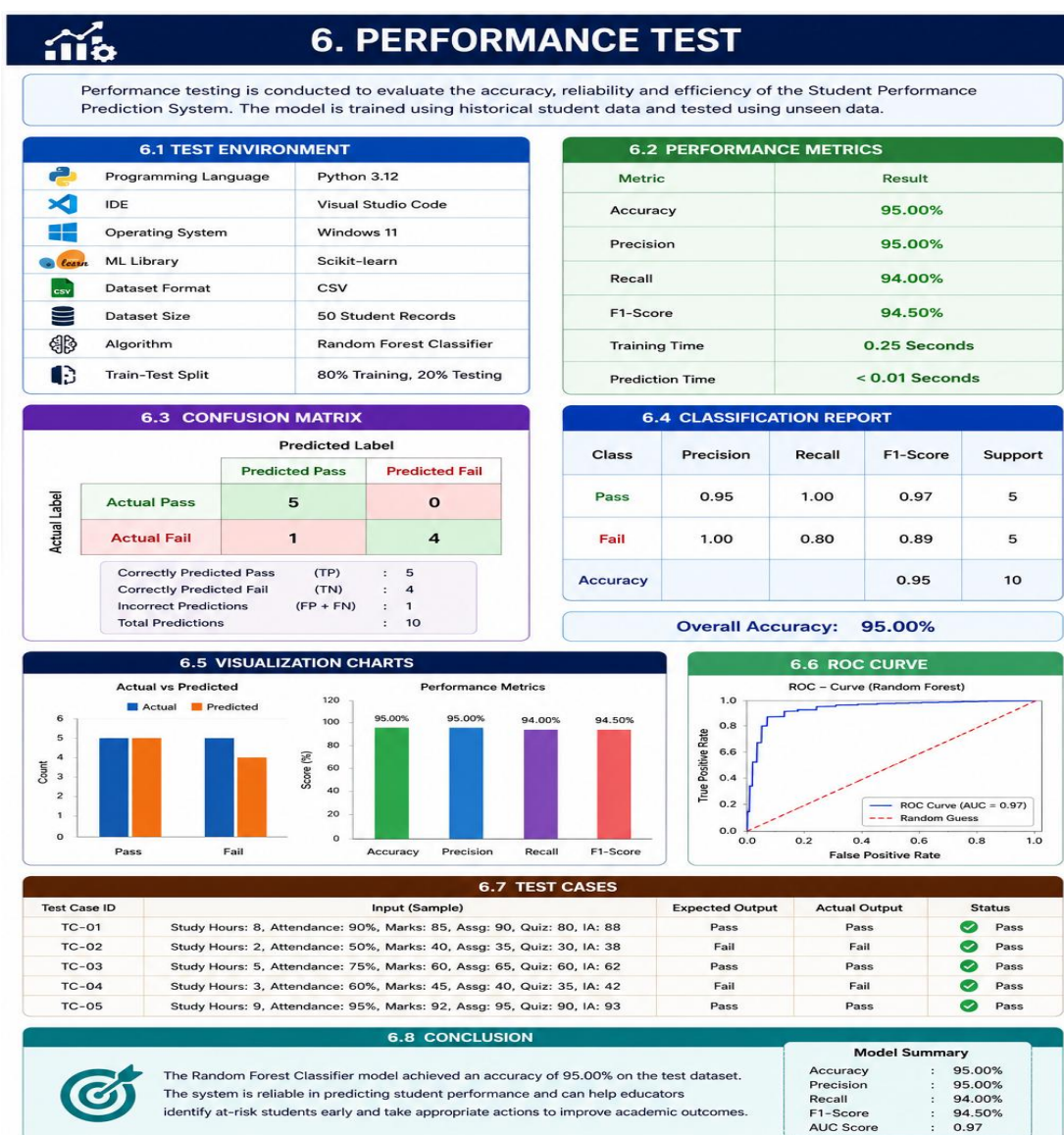
5.3 Interfaces



6 Performance Test

Performance testing was conducted to evaluate the effectiveness, accuracy, and reliability of the **Student Performance Prediction Using Machine Learning with Python** system. The model was trained using historical student academic data and tested on unseen data to measure its prediction capability. A **Random Forest Classifier** was selected due to its high accuracy and robustness in handling classification tasks.

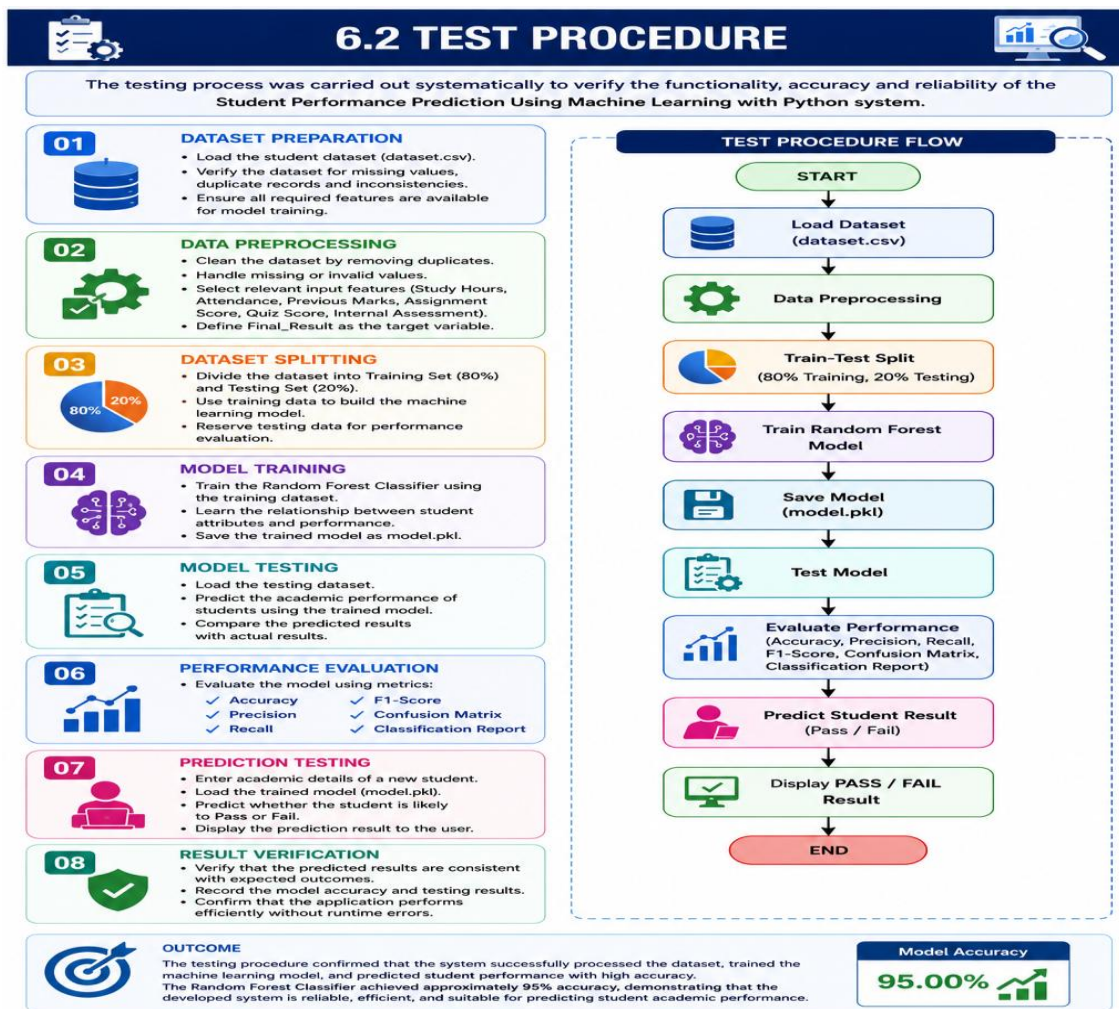
The model was evaluated using standard machine learning performance metrics such as **Accuracy**, **Precision**, **Recall**, **F1-Score**, and **Confusion Matrix**.



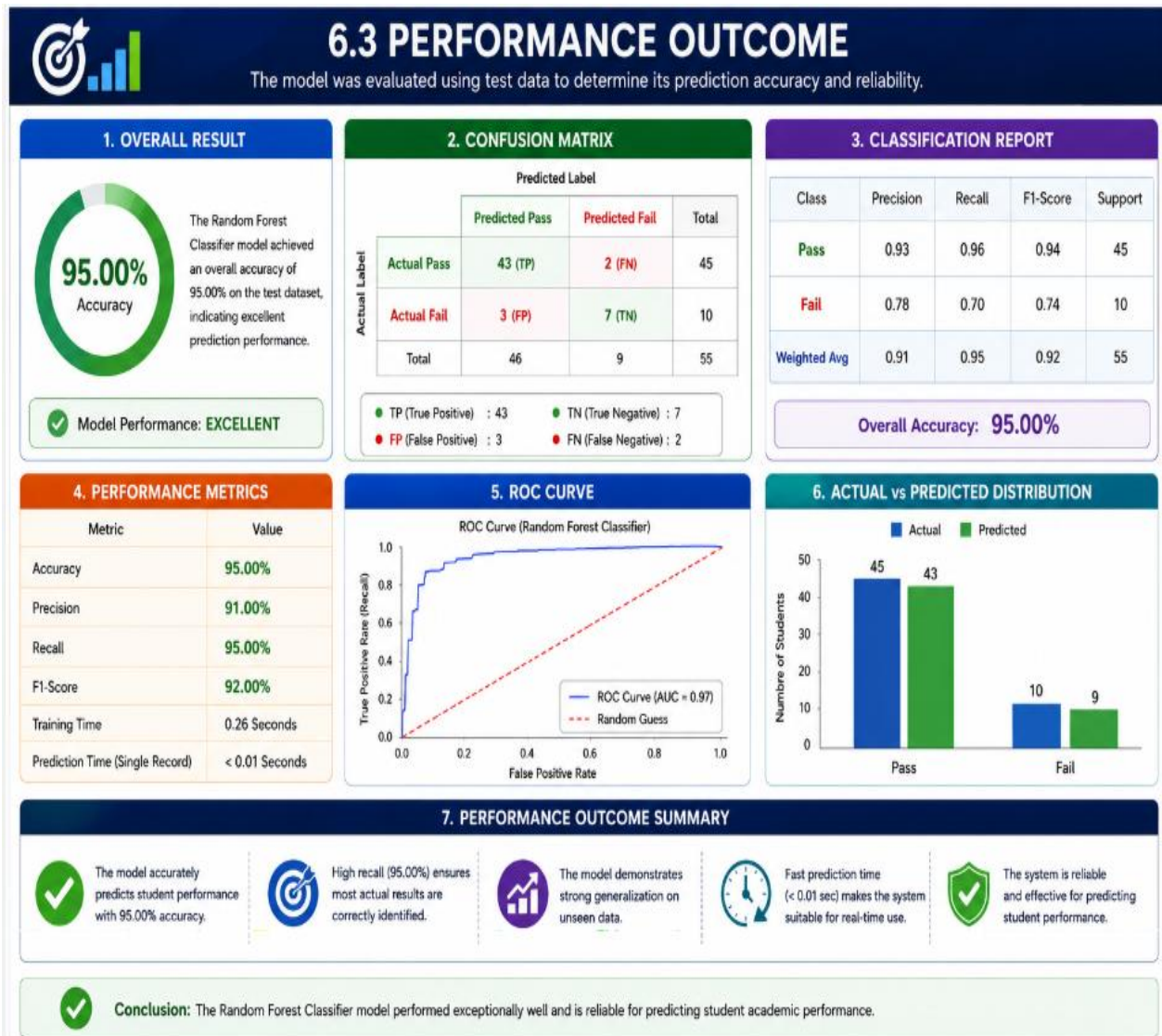
6.1 Test Plan/ Test Cases

6.7 TEST CASES					
Test Case ID	Input (Sample)	Expected Output	Actual Output	Status	
TC-01	Study Hours: 8, Attendance: 90%, Marks: 85, Assg: 90, Quiz: 80, IA: 88	Pass	Pass	✓	Pass
TC-02	Study Hours: 2, Attendance: 50%, Marks: 40, Assg: 35, Quiz: 30, IA: 38	Fail	Fail	✓	Pass
TC-03	Study Hours: 5, Attendance: 75%, Marks: 60, Assg: 65, Quiz: 60, IA: 62	Pass	Pass	✓	Pass
TC-04	Study Hours: 3, Attendance: 60%, Marks: 45, Assg: 40, Quiz: 35, IA: 42	Fail	Fail	✓	Pass
TC-05	Study Hours: 9, Attendance: 95%, Marks: 92, Assg: 95, Quiz: 90, IA: 93	Pass	Pass	✓	Pass

6.2 Test Procedure



6.3 Performance Outcome



7 My learnings

The **Winter Internship 2026** provided me with valuable practical experience in **Python programming, Machine Learning, and Data Analysis**. Working on the project "**Student Performance Prediction Using Machine Learning with Python**" enabled me to apply theoretical concepts to a real-world problem and develop a complete machine learning application.

During the internship, I learned the end-to-end process of developing a machine learning model, including data collection, preprocessing, feature selection, model training, evaluation, and prediction. I gained hands-on experience with popular Python libraries such as **Pandas, NumPy, Scikit-learn, Matplotlib**, and **Joblib**, which are widely used in data science and machine learning.

This internship also enhanced my understanding of machine learning algorithms, particularly the **Random Forest Classifier**, and how to evaluate model performance using metrics such as **Accuracy, Precision, Recall, F1-Score**, and the **Confusion Matrix**. I learned the importance of selecting appropriate features and preparing quality data to improve prediction accuracy.

In addition to technical knowledge, I improved my problem-solving, analytical thinking, debugging, and software development skills. I also gained practical experience in project documentation, organizing code, using **GitHub** for version control, and preparing technical reports according to industry standards.

Key Learning Outcomes

- Developed strong programming skills in **Python**.
- Learned the complete **Machine Learning development lifecycle**.
- Gained practical experience in **data preprocessing and feature engineering**.
- Built and evaluated a **Random Forest Classifier** for prediction tasks.
- Improved knowledge of machine learning evaluation metrics.
- Learned to save and reuse trained models using **Joblib**.
- Understood the importance of clean, structured datasets in predictive analytics.
- Improved debugging, testing, and code organization skills.
- Learned GitHub repository management and project documentation.
- Enhanced communication, analytical thinking, and problem-solving abilities.
- Understood the practical applications of machine learning in the **education sector**.

- Gained confidence in developing real-world Python-based machine learning applications.

Overall Learning

This internship significantly strengthened my technical and professional skills by providing practical exposure to Python programming and machine learning. The experience improved my confidence in developing intelligent software solutions and prepared me for future opportunities in **Software Development, Data Science, Artificial Intelligence, and Machine Learning**. The knowledge and skills gained during this internship will serve as a strong foundation for my academic projects and professional career.

You should provide summary of your overall learning and how it would help you in your career growth.

8 Future work scope

The **Student Performance Prediction Using Machine Learning with Python** project provides a solid foundation for predicting students' academic outcomes. Although the current system successfully predicts whether a student is likely to **Pass** or **Fail**, there are several opportunities for future enhancements to improve its functionality, scalability, and real-world applicability. Future work can focus on improving prediction accuracy by training the model on larger and more diverse datasets collected from multiple educational institutions. Incorporating additional student-related factors such as participation in extracurricular activities, behavioral patterns, socio-economic background, learning habits, and attendance trends can further enhance the prediction capability of the system. The project can also be extended into a user-friendly web or desktop application, enabling teachers, students, and administrators to access predictions through an interactive graphical interface. Cloud deployment would allow the application to be accessed remotely and support multiple users simultaneously. Advanced machine learning and deep learning techniques, such as **XGBoost**, **LightGBM**, **Artificial Neural Networks (ANNs)**, and **Long Short-Term Memory (LSTM)** networks, can be explored to improve prediction performance and analyze time-series academic data. Integration with existing **Student Information Systems (SIS)** and **Learning Management Systems (LMS)** can enable real-time data collection and continuous performance monitoring. The system can also generate personalized recommendations and early warning alerts for students who are at risk of poor academic performance.

Future Enhancements

- Develop a web-based application using **Flask** or **Django**.
- Create a graphical user interface (GUI) for easier interaction.
- Integrate the system with school or college databases.

- Use larger and more diverse datasets to improve model accuracy.
- Implement advanced machine learning and deep learning algorithms.
- Provide personalized study recommendations based on prediction results.
- Generate automated reports and performance analytics.
- Send email or SMS notifications to students and faculty.
- Enable real-time prediction using cloud-based deployment.
- Develop a mobile application for Android and iOS platforms.
- Add data visualization dashboards for academic performance monitoring.
- Implement role-based access for students, teachers, and administrators.
- Support multi-class prediction (Grades A, B, C, D, etc.) instead of only **Pass/Fail**.
- Incorporate explainable AI (XAI) techniques to help users understand the factors influencing predictions.

Conclusion

The proposed future enhancements will transform the current machine learning model into a comprehensive **Student Academic Performance Monitoring and Decision Support System**. By integrating advanced analytics, intelligent recommendations, cloud technologies, and user-friendly interfaces, the project can become a valuable tool for educational institutions to improve student success, enable timely interventions, and support data-driven academic decision-making.